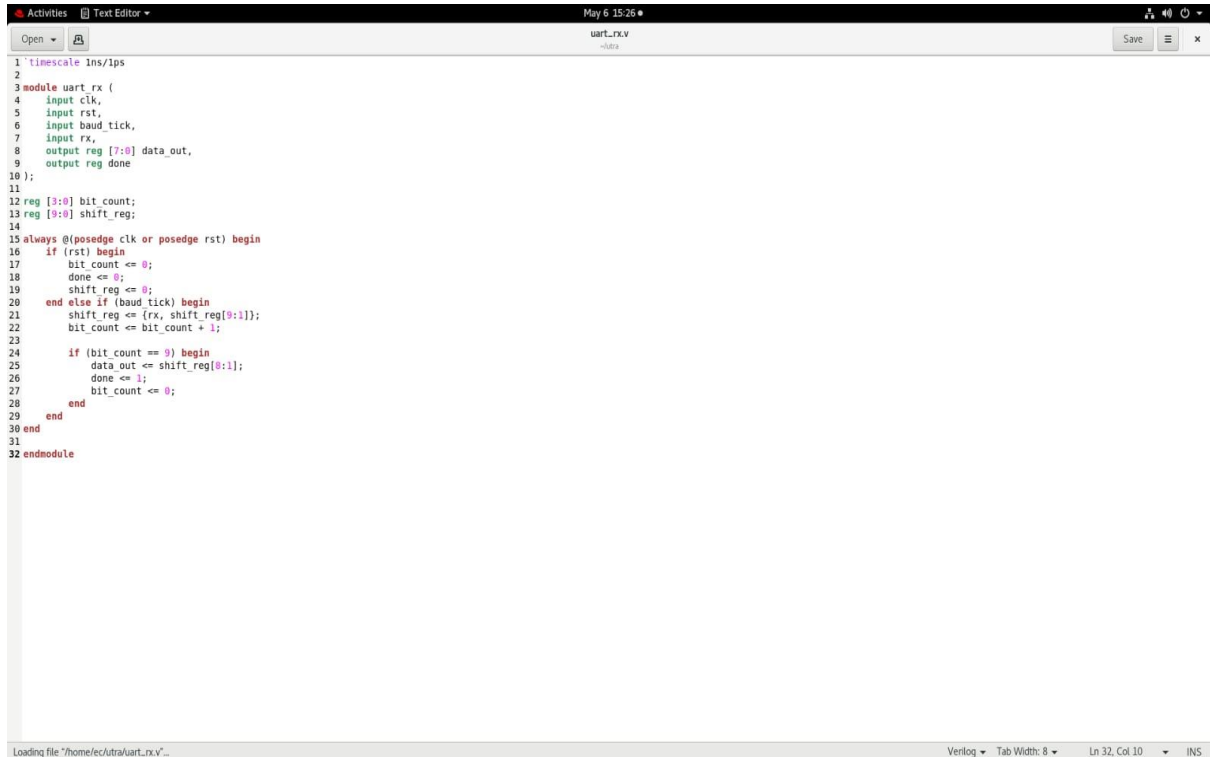


UART_Project_Screenshots

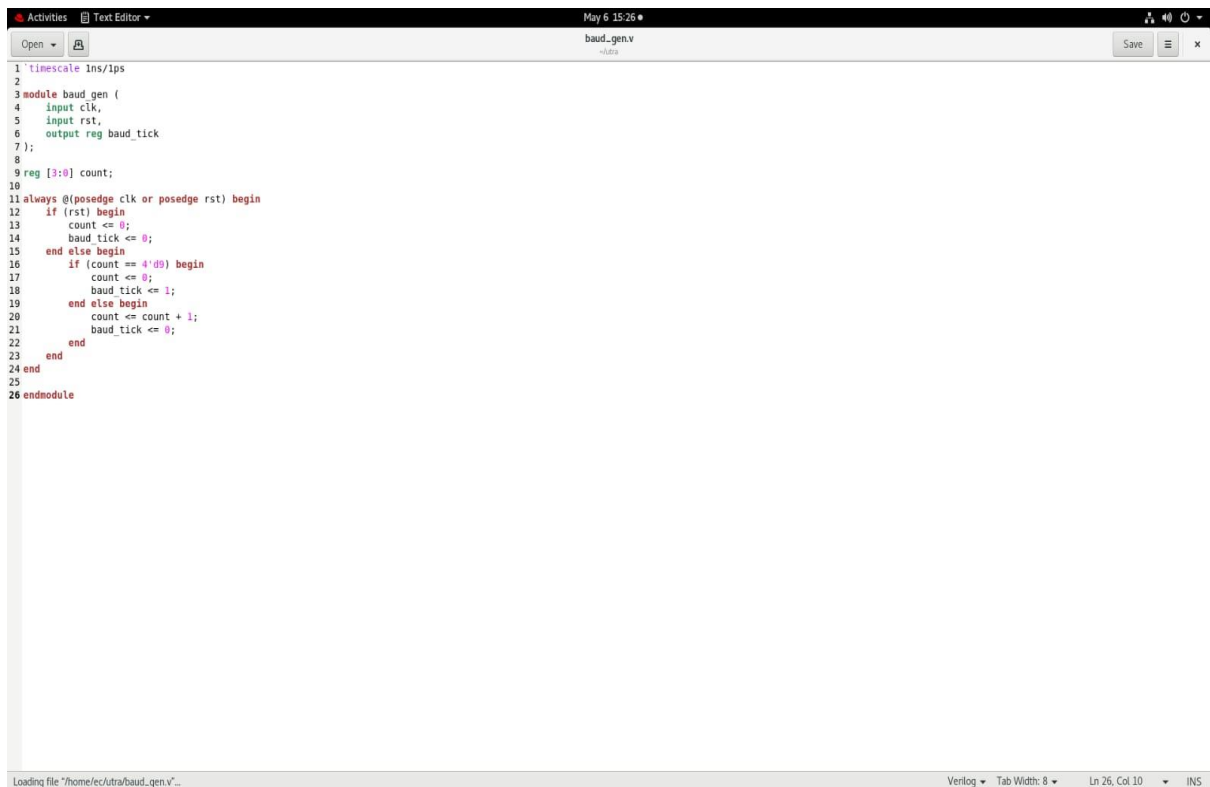


The screenshot shows a Verilog code editor window titled "uart_rx.v". The code defines a module named `uart_rx` with inputs `clk`, `rst`, and `baud_tick`, and outputs `data_out` and `done`. The module includes a 4-bit `bit_count` register and a 9-bit `shift_reg` register. The logic is as follows:

```
1 `timescale 1ns/1ps
2
3 module uart_rx (
4     input clk,
5     input rst,
6     input baud_tick,
7     input rx,
8     output reg [7:0] data_out,
9     output reg done
10 );
11
12 reg [3:0] bit_count;
13 reg [9:0] shift_reg;
14
15 always @(posedge clk or posedge rst) begin
16     if (rst) begin
17         bit_count <= 0;
18         done <= 0;
19         shift_reg <= 0;
20     end else if (baud_tick) begin
21         shift_reg <= {rx, shift_reg[9:1]};
22         bit_count <= bit_count + 1;
23     end
24     if (bit_count == 0) begin
25         data_out <= shift_reg[8:1];
26         done <= 1;
27         bit_count <= 0;
28     end
29 end
30 end
31
32 endmodule
```

The status bar at the bottom indicates the file path: "Loading file "/home/ec/ultra/uart_rx.v"...". The editor settings show "Verilog", "Tab Width: 8", "Ln 32, Col 10", and "INS" mode.

Figure:1

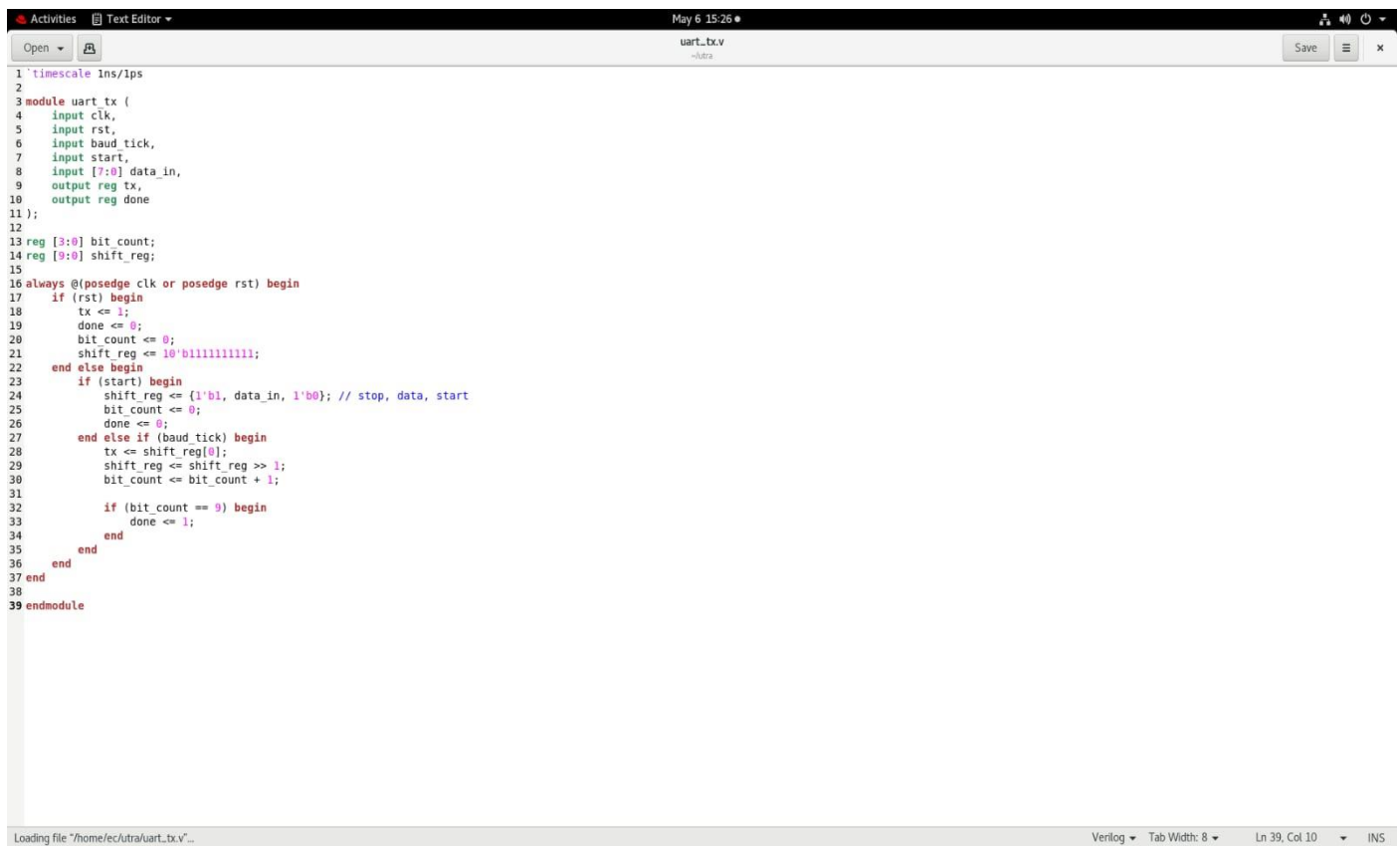


The screenshot shows a Verilog code editor window titled "baud_gen.v". The code defines a module named `baud_gen` with inputs `clk` and `rst`, and an output `baud_tick`. The module includes a 4-bit `count` register. The logic is as follows:

```
1 `timescale 1ns/1ps
2
3 module baud_gen (
4     input clk,
5     input rst,
6     output reg baud_tick
7 );
8
9 reg [3:0] count;
10
11 always @(posedge clk or posedge rst) begin
12     if (rst) begin
13         count <= 0;
14         baud_tick <= 0;
15     end else begin
16         if (count == 4'd9) begin
17             count <= 0;
18             baud_tick <= 1;
19         end else begin
20             count <= count + 1;
21             baud_tick <= 0;
22         end
23     end
24 end
25
26 endmodule
```

The status bar at the bottom indicates the file path: "Loading file "/home/ec/ultra/baud_gen.v"...". The editor settings show "Verilog", "Tab Width: 8", "Ln 26, Col 10", and "INS" mode.

Figure:2

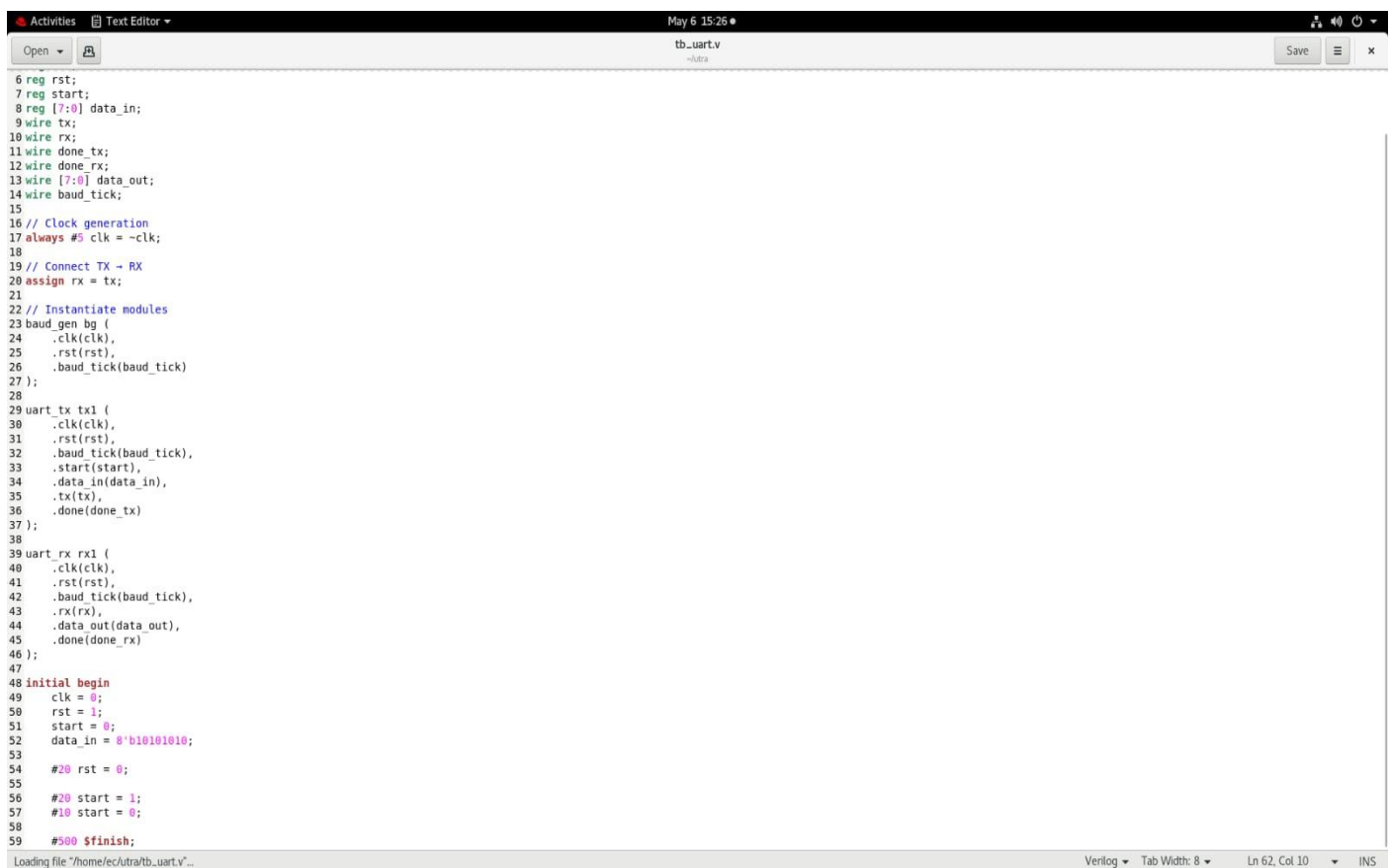


The screenshot shows a text editor window titled 'uart_tx.v' with the following Verilog code:

```
1 `timescale 1ns/1ps
2
3 module uart_tx (
4     input clk,
5     input rst,
6     input baud_tick,
7     input start,
8     input [7:0] data_in,
9     output reg tx,
10    output reg done
11);
12
13 reg [3:0] bit_count;
14 reg [9:0] shift_reg;
15
16 always @(posedge clk or posedge rst) begin
17     if (rst) begin
18         tx <= 1;
19         done <= 0;
20         bit_count <= 0;
21         shift_reg <= 10'b1111111111;
22     end else begin
23         if (start) begin
24             shift_reg <= {1'b1, data_in, 1'b0}; // stop, data, start
25             bit_count <= 0;
26             done <= 0;
27         end else if (baud_tick) begin
28             tx <= shift_reg[0];
29             shift_reg <= shift_reg >> 1;
30             bit_count <= bit_count + 1;
31
32             if (bit_count == 9) begin
33                 done <= 1;
34             end
35         end
36     end
37 end
38
39 endmodule
```

The status bar at the bottom indicates the file path is 'Loading file "/home/ec/ultra/uart_tx.v"...', the editor is in Verilog mode, tab width is 8, and the cursor is at line 39, column 10.

Figure:3

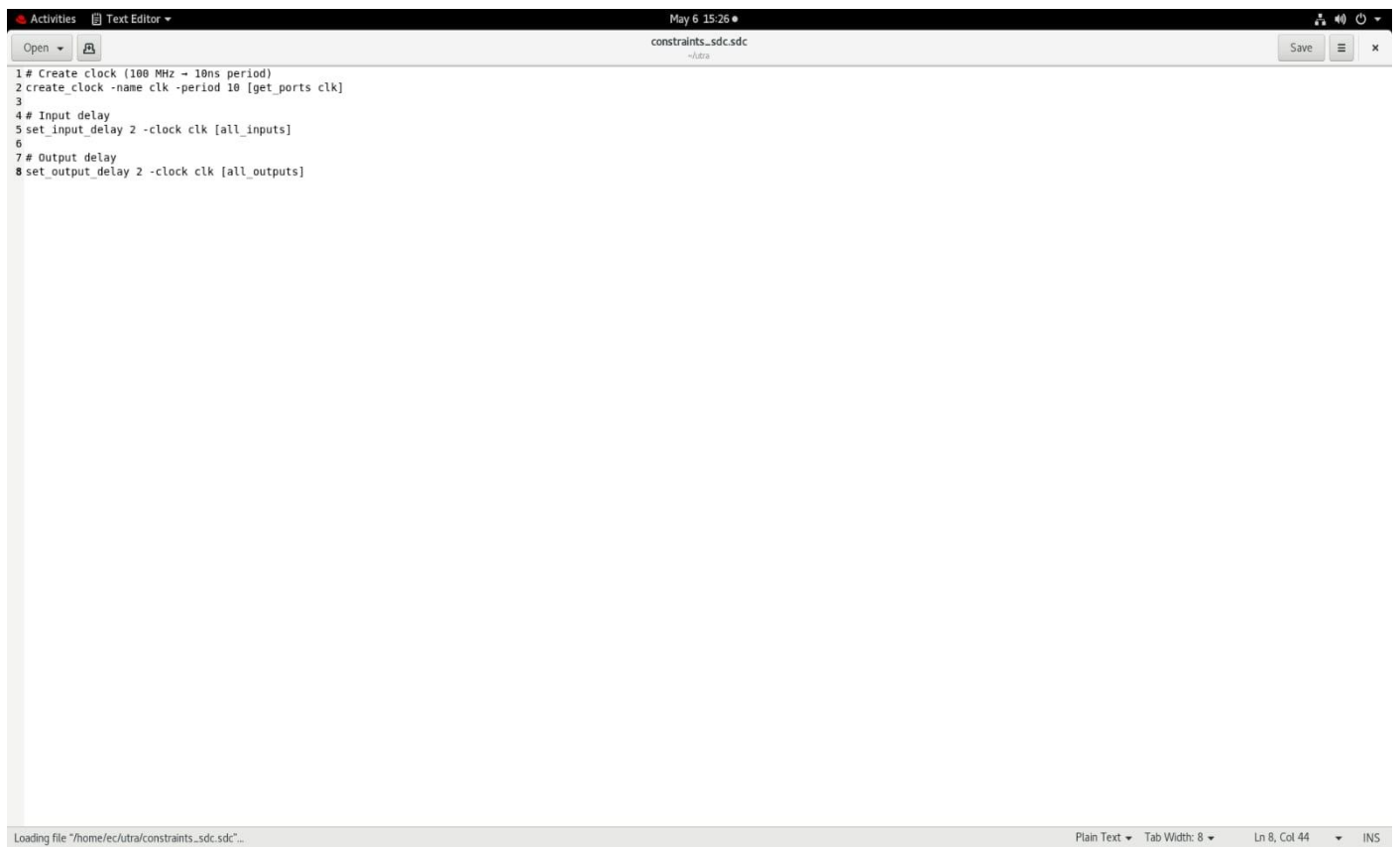


The screenshot shows a text editor window titled 'tb_uart.v' with the following Verilog code:

```
6 reg rst;
7 reg start;
8 reg [7:0] data_in;
9 wire tx;
10 wire rx;
11 wire done_tx;
12 wire done_rx;
13 wire [7:0] data_out;
14 wire baud_tick;
15
16 // Clock generation
17 always #5 clk = ~clk;
18
19 // Connect TX ~ RX
20 assign rx = tx;
21
22 // Instantiate modules
23 baud_gen bg (
24     .clk(clk),
25     .rst(rst),
26     .baud_tick(baud_tick)
27);
28
29 uart_tx tx1 (
30     .clk(clk),
31     .rst(rst),
32     .baud_tick(baud_tick),
33     .start(start),
34     .data_in(data_in),
35     .tx(tx),
36     .done(done_tx)
37);
38
39 uart_rx rx1 (
40     .clk(clk),
41     .rst(rst),
42     .baud_tick(baud_tick),
43     .rx(rx),
44     .data_out(data_out),
45     .done(done_rx)
46);
47
48 initial begin
49     clk = 0;
50     rst = 1;
51     start = 0;
52     data_in = 8'b10101010;
53
54     #20 rst = 0;
55
56     #20 start = 1;
57     #10 start = 0;
58
59     #500 $finish;
60 end
```

The status bar at the bottom indicates the file path is 'Loading file "/home/ec/ultra/tb_uart.v"...', the editor is in Verilog mode, tab width is 8, and the cursor is at line 62, column 10.

Figure:4

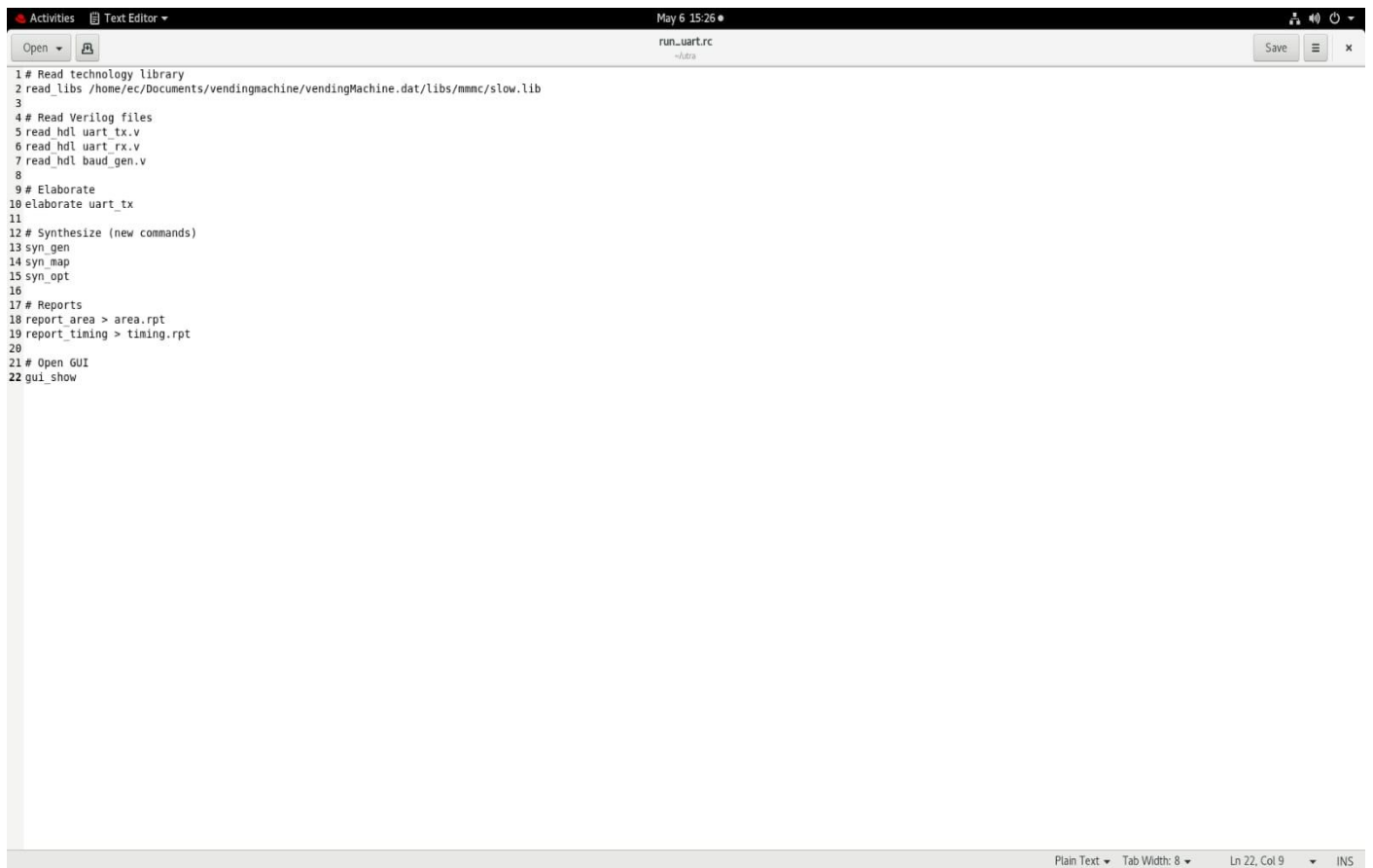


A screenshot of a text editor window titled "Text Editor" with a timestamp of "May 6 15:26". The window displays the content of a file named "constraints.sdc". The code is as follows:

```
1 # Create clock (100 MHz → 10ns period)
2 create_clock -name clk -period 10 [get_ports clk]
3
4 # Input delay
5 set_input_delay 2 -clock clk [all_inputs]
6
7 # Output delay
8 set_output_delay 2 -clock clk [all_outputs]
```

The status bar at the bottom indicates "Loading file "/home/ec/ultra/constraints.sdc"...". On the right side of the status bar, it shows "Plain Text", "Tab Width: 8", "Ln 8, Col 44", and "INS".

Figure:5



A screenshot of a text editor window titled "Text Editor" with a timestamp of "May 6 15:26". The window displays the content of a file named "run_uart.rc". The code is as follows:

```
1 # Read technology library
2 read_libs /home/ec/Documents/vendingmachine/vendingMachine.dat/libs/mmc/slow.lib
3
4 # Read Verilog files
5 read_hdl uart_tx.v
6 read_hdl uart_rx.v
7 read_hdl baud_gen.v
8
9 # Elaborate
10 elaborate uart_tx
11
12 # Synthesize (new commands)
13 syn_gen
14 syn_map
15 syn_opt
16
17 # Reports
18 report_area > area.rpt
19 report_timing > timing.rpt
20
21 # Open GUI
22 gui_show
```

The status bar at the bottom indicates "Plain Text", "Tab Width: 8", "Ln 22, Col 9", and "INS".

Figure:6

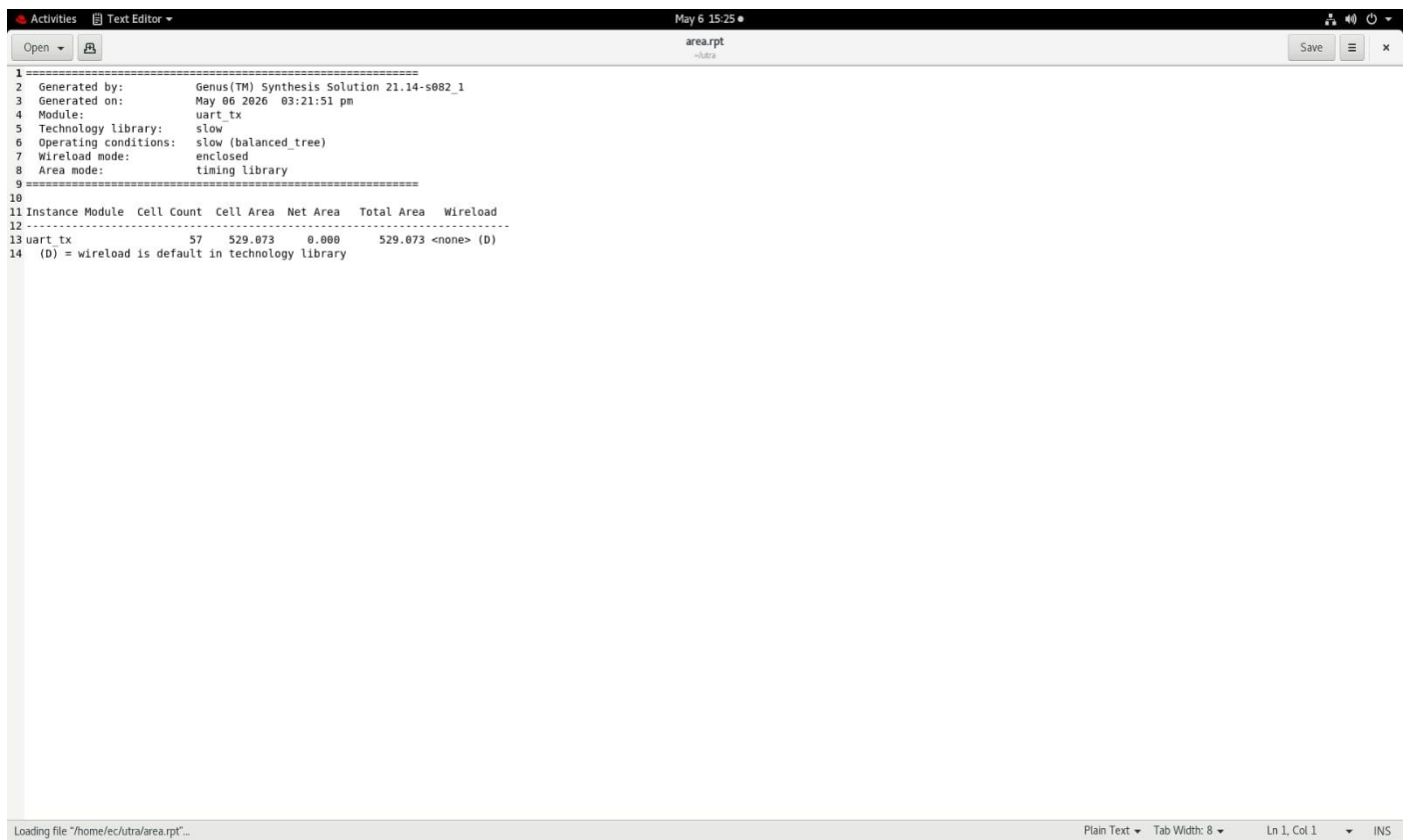


Figure:7

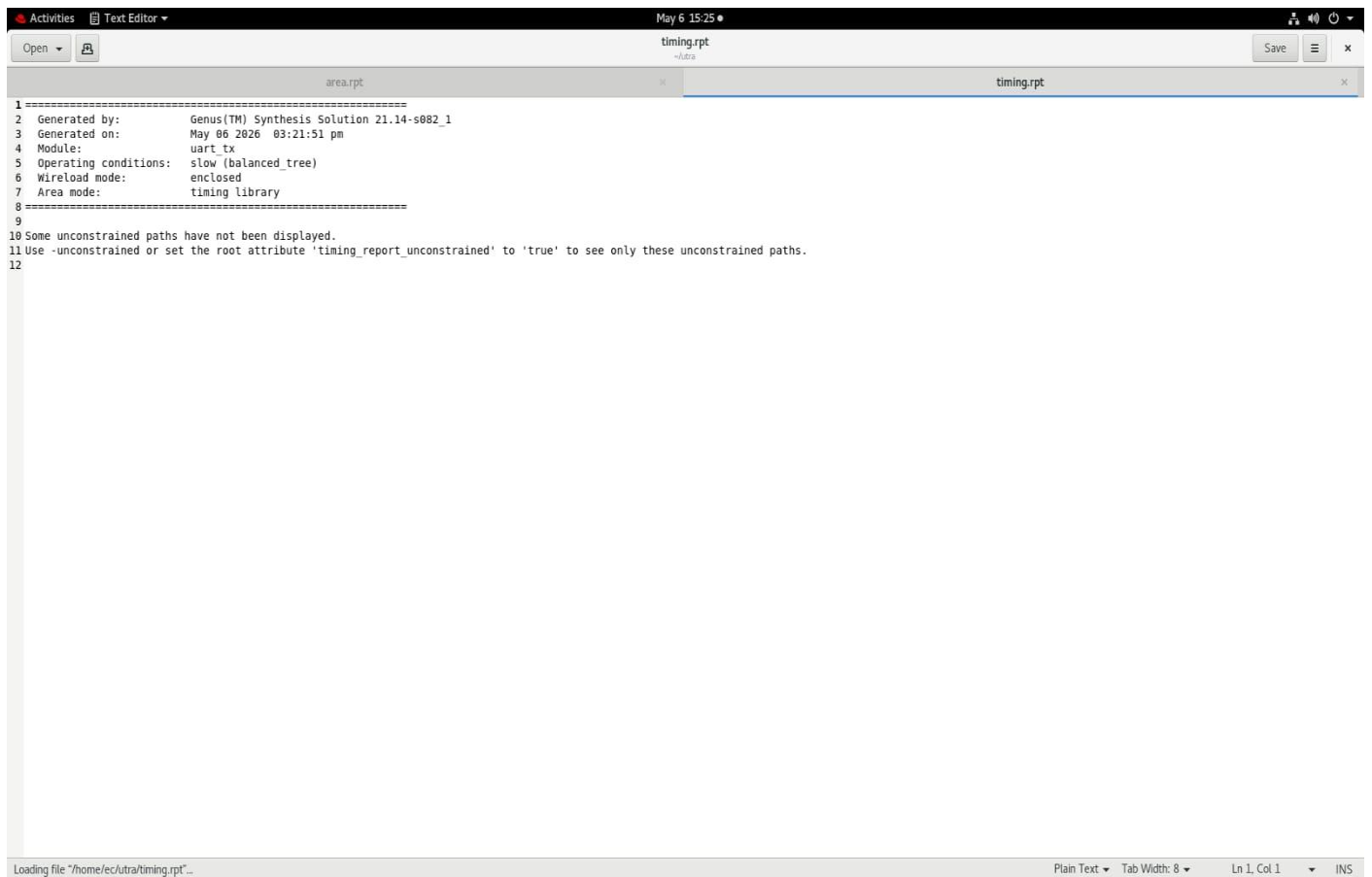


Figure:8

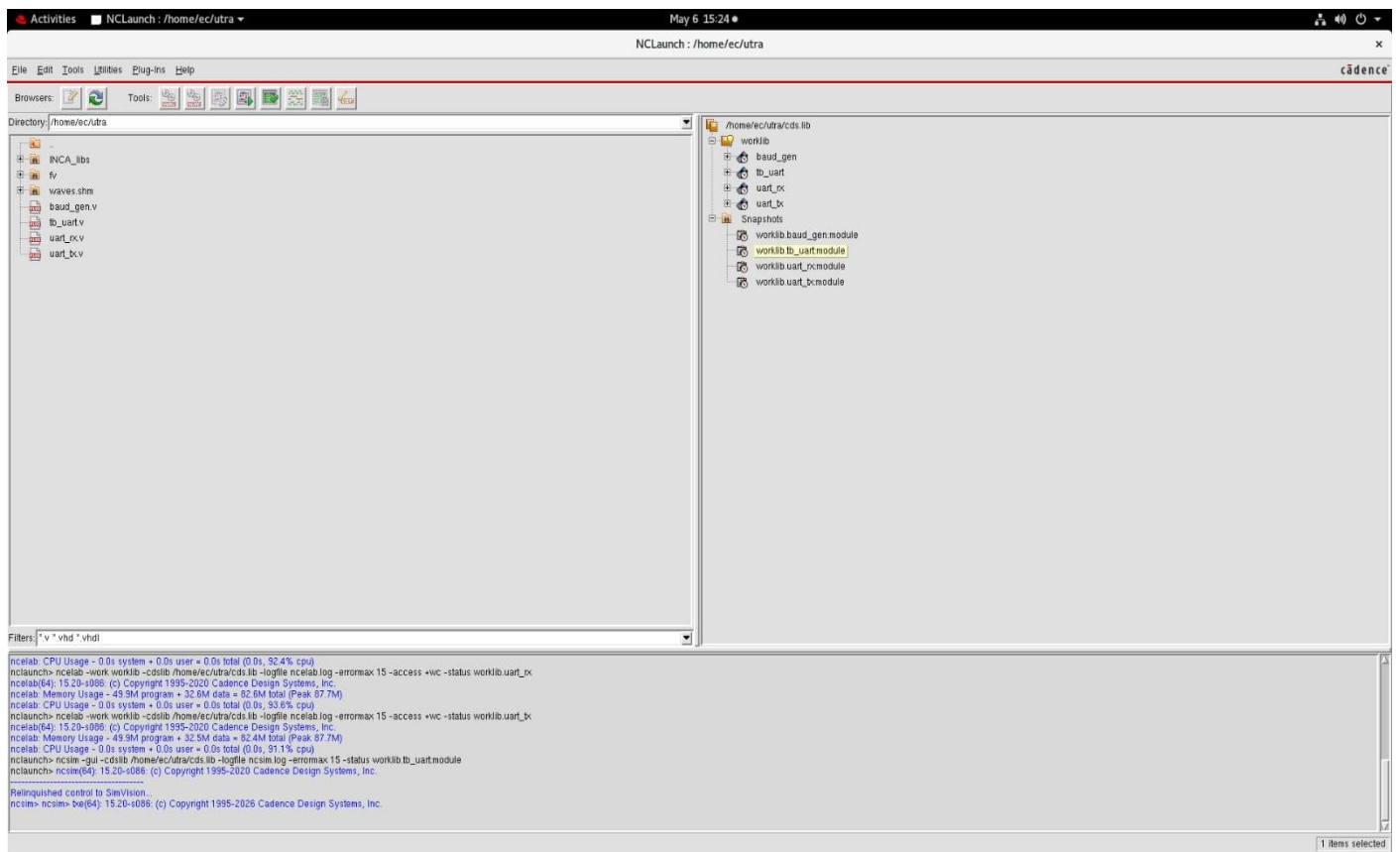


Figure:9

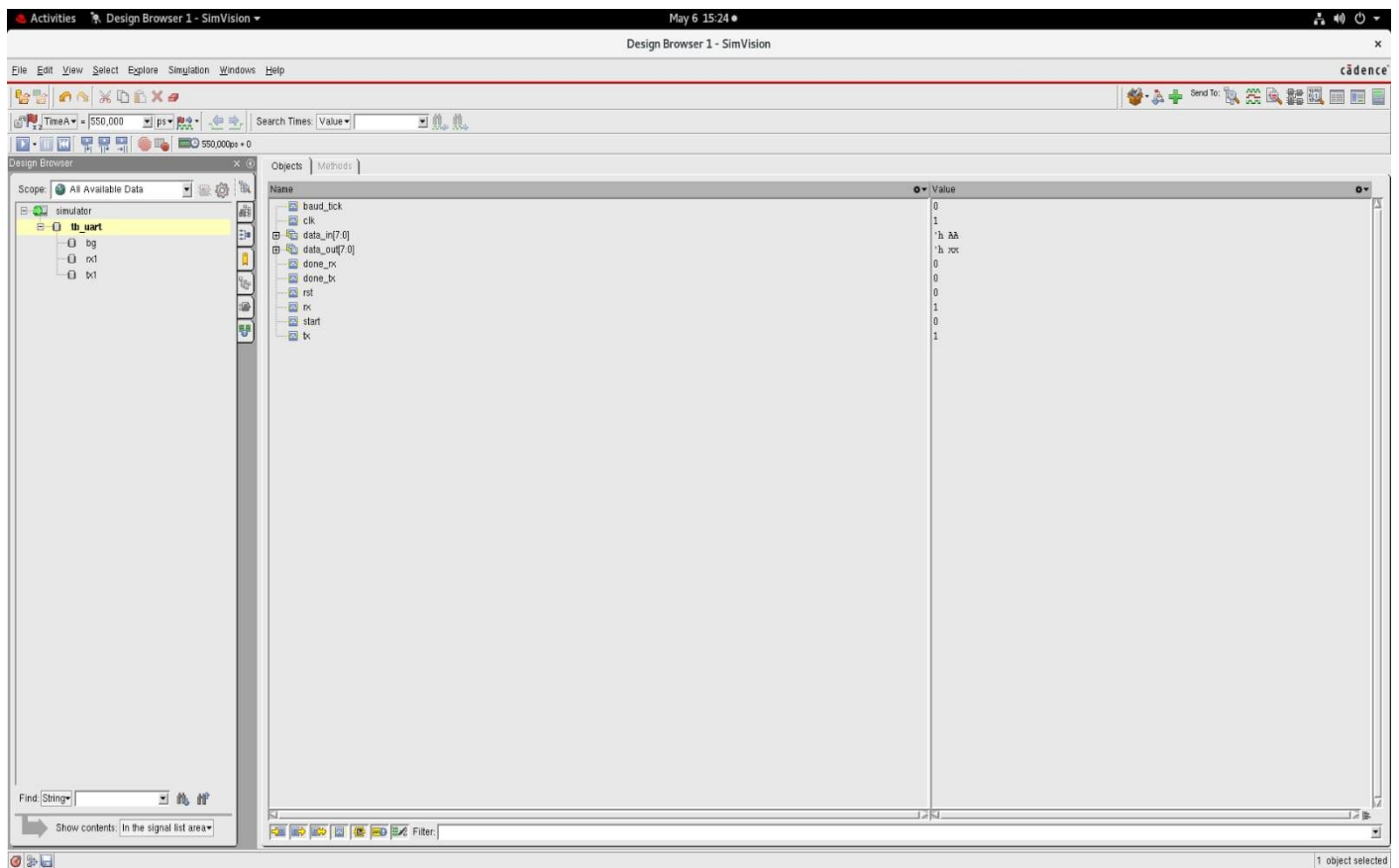


Figure:10

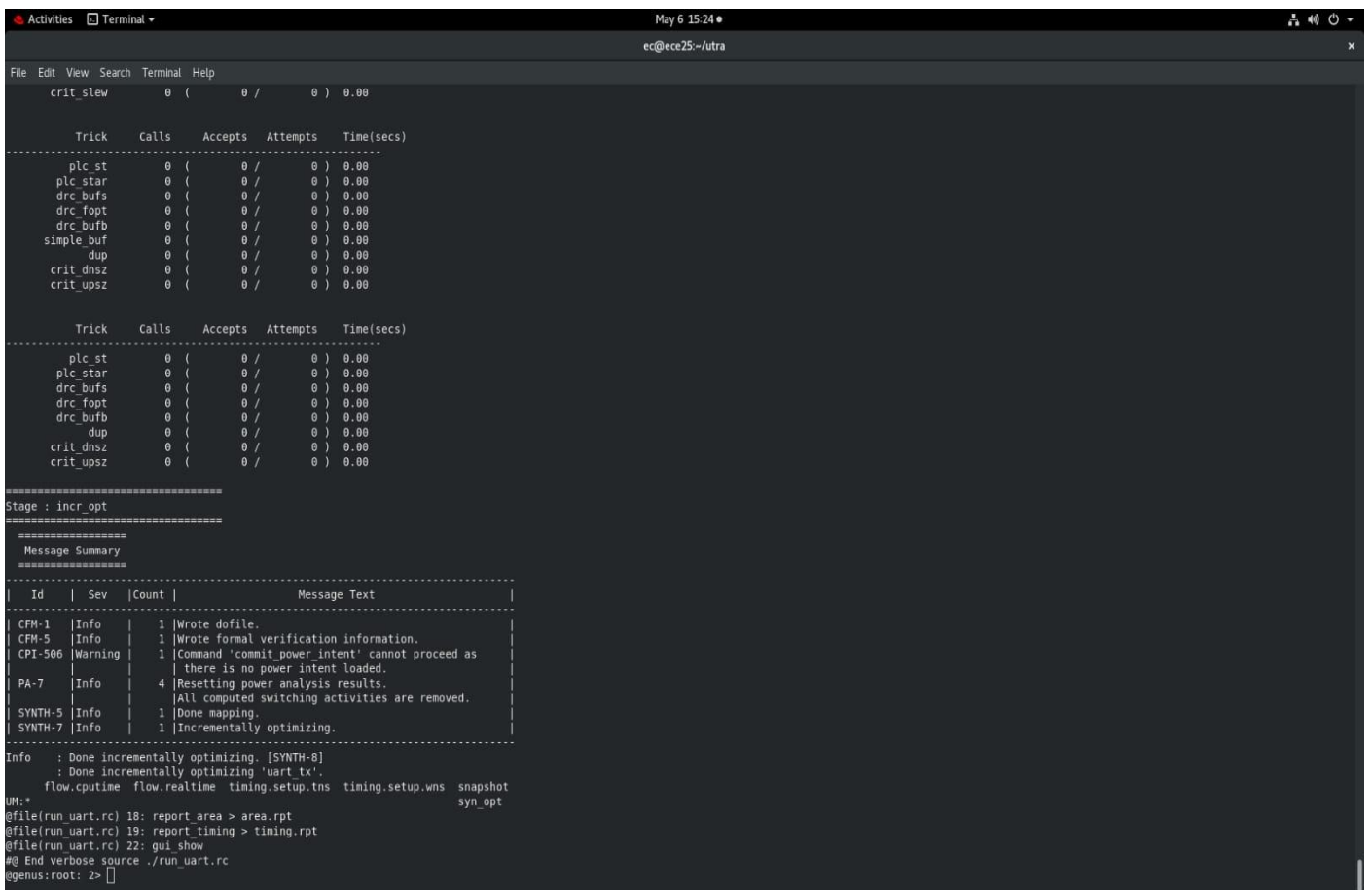


Figure:11

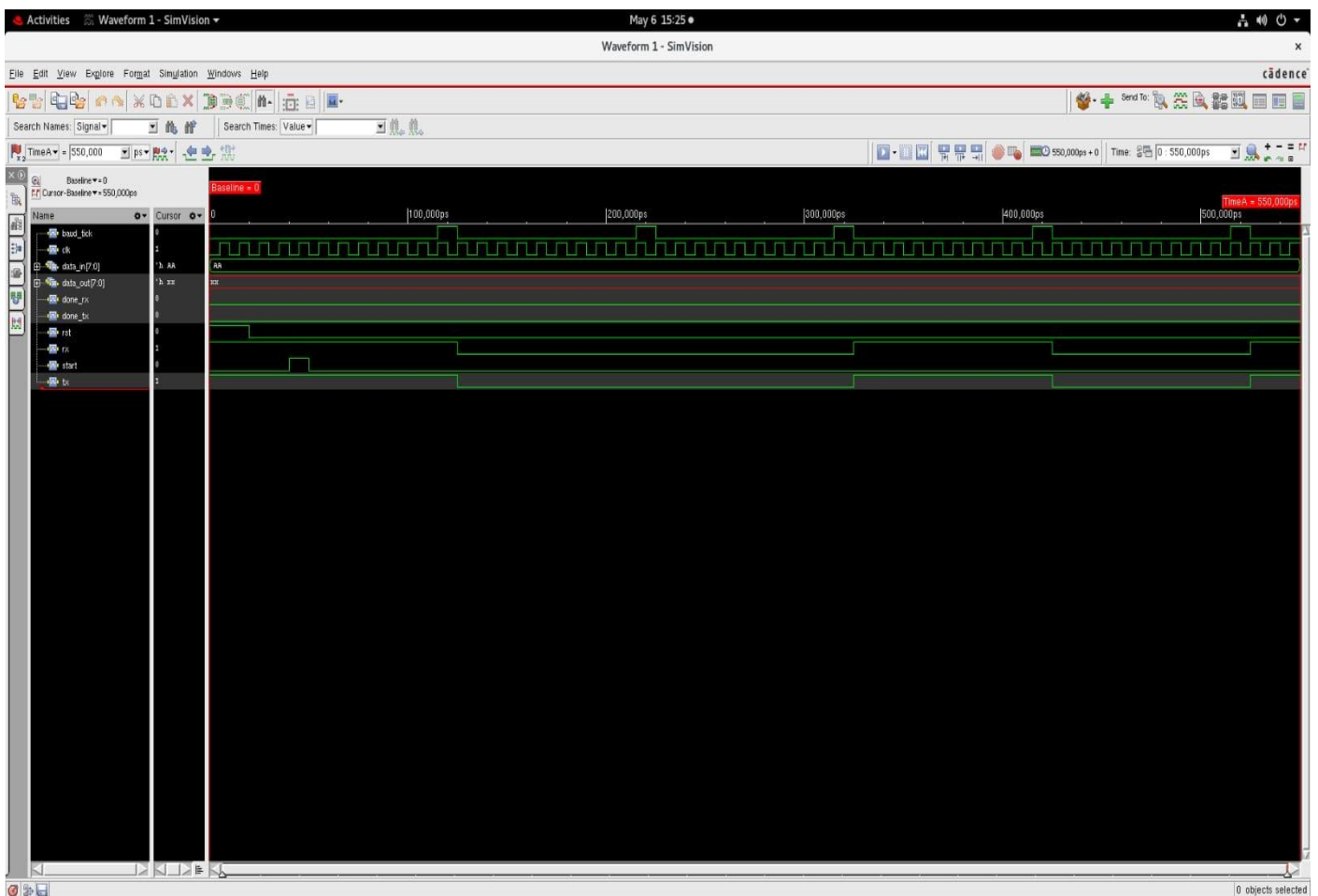


Figure:12

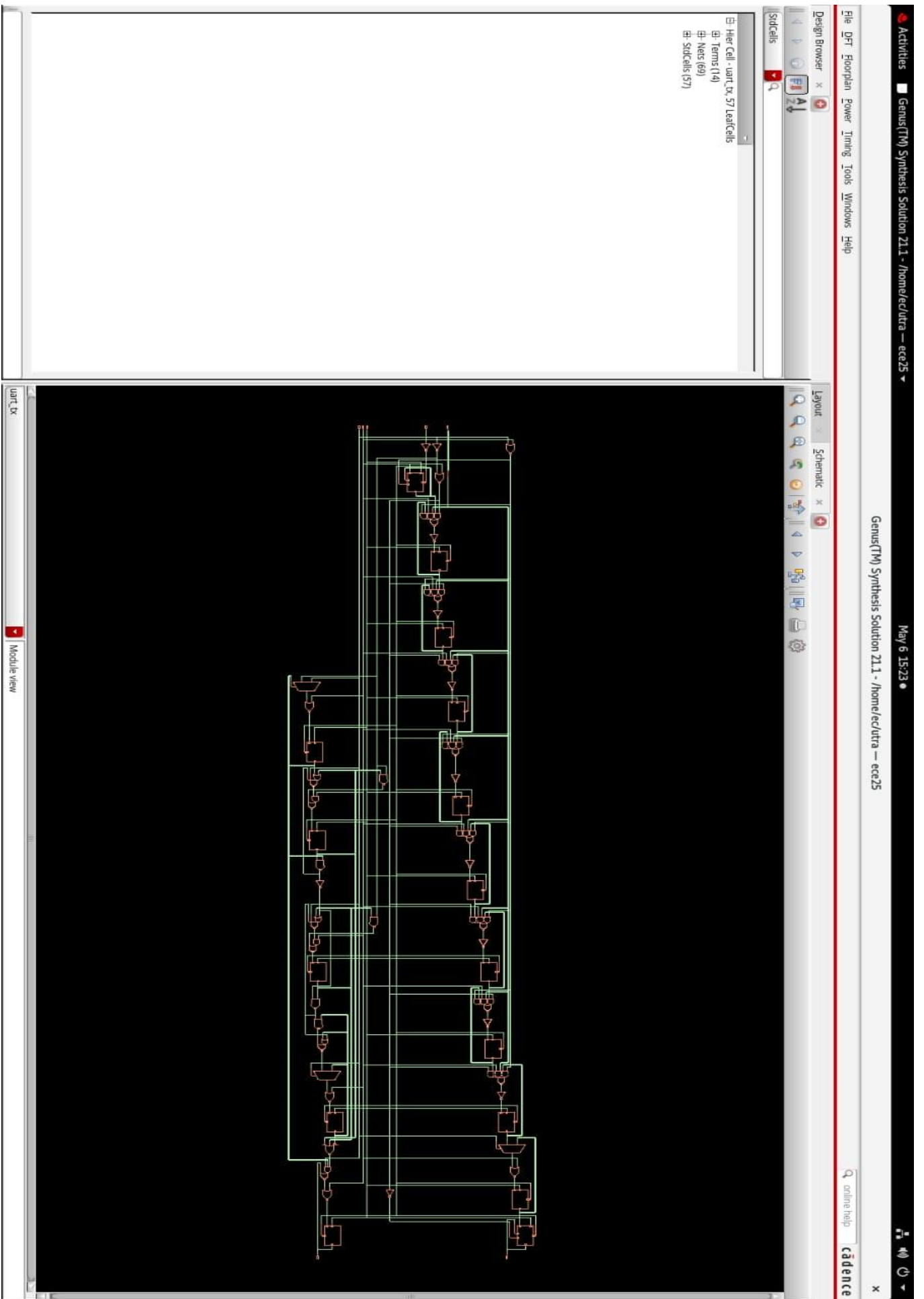


Figure:13